

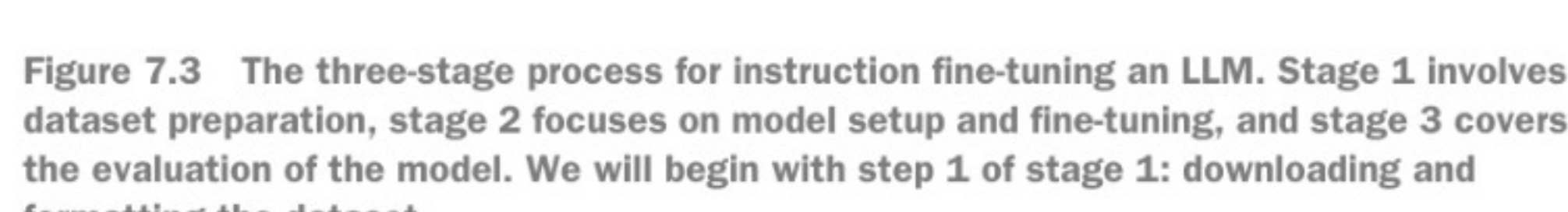
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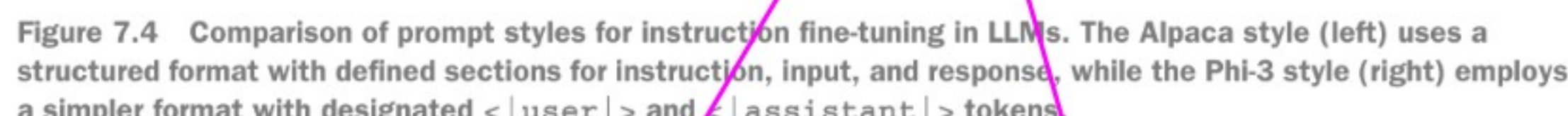
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- The diagram illustrates the process of generating a response from an LLM based on instructions. It is organized into two main columns: 'Instruction' and 'Desired response'. The 'Instruction' column is headed by the text 'The instructions serve as inputs for the LLM.' with an arrow pointing to the 'Instruction' label. The 'Desired response' column is headed by the text 'The goal for the LLM is to generate a desired response.' with an arrow pointing to the 'Desired response' label. Three examples are provided, each consisting of an instruction, a right-pointing arrow, and the corresponding desired response.
- | Instruction | Desired response |
|---|--------------------------------------|
| Convert 45 kilometers to meters. | 45 kilometers is 45000 meters. |
| Provide a synonym for "bright." | A synonym for "bright" is "radiant." |
| Edit the following sentence to remove all passive voice: "The song was composed by the artist." | The artist composed the song. |

Figure 7.2 Examples of instructions that are processed by an LLM to generate desired responses

- #### 4. Instruction fine-tuning workflow



- * How to format input-output pairs.



Phi-3 prompt style.

- * partitioning the dataset

- (Train - Test - validation)

↓	↓	↓
85%	10%	5%